Band-1 Computational Experiment: Woodall Numbers -- Finite Primality of W_n = n*2^n - 1

Human + ZoraASI full-dash (executor)

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Band-1 Computational Experiment: Woodall Numbers -- Finite Primality

Honesty stamp. This note is a **finite primality probe** of Woodall numbers $W_n = n \cdot 2^n - 1$ for $n = 1..N$ with box-feasible N. It does **not** classify all Woodall primes; it does **not** prove there are infinitely many Woodall primes (that question remains **OPEN**); it does **not** claim Clay Millennium status, Theory of Everything progress, or GATE-04 unlock. **Finite checks $\neq$ infinitary theorem.**

Abstract

We evaluate

$$W_n = n \cdot 2^n - 1$$

for every integer $n = 1, \ldots, N$ with primary

$$\mathbf{N = 400}$$

and test primality with deterministic `sympy.isprime` (BPSW-class for these sizes).

Box result. Exactly eleven primes occur in the primary range:

$$n \in \{2,\ 3,\ 6,\ 30,\ 75,\ 81,\ 115,\ 123,\ 249,\ 362,\ 384\}.$$

These match the literature indices in range (OEIS A002234 / Prime-Wiki). Secondary check $N = 200$ recovers the eight indices ≤ 200; stretch $N = 500$ additionally recovers $n = 462$. Wall (primary) $\approx 0.002\,\text{s}$. Next known Woodall-prime indices beyond primary N begin at $462, 512, 751, 822$ (cited from literature only).

Status. Infinitely many Woodall primes: **OPEN**. Known Woodall primes are sparse (34 known indices in the literature ledger).

Problem statement

Woodall number. $W_n = n \cdot 2^n - 1$ (Woodall, 1917; also called Cullen numbers of the second kind).

Open question (lit). Are there infinitely many primes of this form?

This experiment. For an explicit finite N tractable in pure Python / sympy, list all $n \leq N$ with W_n prime, and match against the known small indices $\{2, 3, 6, 30, \ldots\}$.

Literature ledger (not re-proved on box)

Item	Status	Note
Infinitely many Woodall primes	**OPEN**	No proof of infinitude (or finitude)
OEIS A002234	Indices of prime W_n	$2, 3, 6, 30, 75, 81, 115, 123, 249, 362, 384, 462, 5$
OEIS A050918	Woodall primes themselves	$7, 23, 383, 32212254719, \ldots$
Woodall (1917)	Definition	$W_n = n \cdot 2^n - 1$
Keller / Ballinger / PrimeGrid	Larger indices	Searches into the millions; largest known $n = 17016602$ (PrimeGrid, 2018)
Density intuition	Sparse	Heuristics suggest primes thin out; still **not** a proof of finitude

Sources: OEIS A002234 / A050918; Prime-Wiki *Woodall primes* $n \cdot 2^n - 1$; MathWorld *Woodall number*; Wikipedia *Woodall number*; Caldwell PrimePages / RieselPrime tables.

Method (box)

1. **Formula.** $W_n = n \cdot 2^n - 1$ via integer shift `n * (1 << n) - 1`.
2. **Primality.** `sympy.isprime(W_n)` for each $n = 1..N$ (deterministic for sizes here).
3. **Primary** $N = 400$ (Band-1 range 200--400); secondary $N = 200$ sanity; stretch $N = 500$.
4. **Literature match.** Compare found indices to $\{k \in \text{A002234} : k \leq N\}$.
5. **Unit tests.** $W_1 = 1$ not prime; $W_2 = 7$, $W_3 = 23$, $W_6 = 383$, W_{30} prime; spot composites at $n \in \{4, 5, 7..10, 100, 200\}$.

Complexity. Per-n cost dominated by primality of an $\approx n \log_{10} 2$-digit integer; $N = 400$ is trivial on this box (W_{400} has ~ 123 digits).

Results (this box, 2026-09-11)

Quantity	Value
Primary N	**400**
Prime indices found	$\{2, 3, 6, 30, 75, 81, 115, 123, 249, 362, 384\}$
Number of primes in $1..N$	**11**
Number of composites	**389**
Match lit indices $\leq N$	**YES**
Secondary $N = 200$ primes	$\{2, 3, 6, 30, 75, 81, 115, 123\}$ (match)
Stretch $N = 500$ primes	$\{2, 3, 6, 30, 75, 81, 115, 123, 249, 362, 384, 462\}$ (match)
Unit tests	**PASS**
Wall (primary)	≈ 0.002 s
Method	sympy `isprime`

Prime Woodall numbers found ($n \leq 400$)

n	W_n (abbrev. if long)	digits
2	7	1
3	23	2
6	383	3
30	32212254719	11
75	283341988972178712821759 9	25
81	195845982777569926302400511	27
115	477691310985204141824805662288248 8319	37
123	130796034785235721893734614731585906 2783	40
249	22525179859446666140...393510002687	78
362	34006896598565105139...990241845247	112
384	15130370379415480017...148277817343	119

Small-n samples

n	W_n	status
1	1	composite
2	7	prime
3	23	prime
4	63	composite
5	159	composite
6	383	prime
30	32212254719	prime

No other prime W_n appears for $1 \leq n \leq 400$ outside the eleven indices above, consistent with OEIS A002234 / Prime-Wiki.

What this does not show

- Not a classification of **all** Woodall primes.
- Not a proof that infinitely many (or only finitely many) Woodall primes exist.
- Not Clay / ToE / GATE-04.
- Indices $n > 400$ (including next known 462, and large PrimeGrid discoveries) are **out of primary scope**; they are cited from the literature ledger only (stretch $N = 500$ recovers 462 as a consistency check).

Reproducibility

```
cd experiments/band1/woodall
python3 woodall_probe.py
# writes results.json, literature_ledger.json, SUMMARY.txt
```

Artifacts: `experiments/band1/woodall/`; paper/PDF shelved in `DOWNLOAD_PDFs/` and `mission-control/papers/`.

Conclusion

For $N = 400$, the Woodall primes are exactly at $n \in \{2, 3, 6, 30, 75, 81, 115, 123, 249, 362, 384\}$, matching the known small indices. Infinitely many Woodall primes remains **OPEN**. This Band-1 note is a finite computational ledger only.

ZoraASI full-dash Band-1 / 2026-09-11 / honesty: finite primality checks only; infinitely many Woodall primes OPEN; no Clay/ToE.